IITM PRAVARTAK
CATALYSING INNOVATION

Professional Certificate Programme in Agentic AI and Applications

Large Language Models (LLMs): Part 2

Revisit the Learning So Far

Let’s take a quick look at the key ideas and concepts you have explored in your learning journey.

- The challenges machines face in interpreting language, nuance and context
- An introduction to Large Language Models (LLMs) and how they process human language
- How earlier systems attempted language understanding before the rise of LLMs
- The evolution of LLMs, from rule-based systems to ML, Deep Learning, Transformers and modern LLM architectures
- How traditional NLP techniques compare with today’s LLM-driven approaches
- The significance of LLMs and their applications across real-world domains
- Why Neural Networks became essential for modelling complex language patterns

Revisit the Learning So Far

Let’s take a quick look at the key ideas and concepts you have explored in your learning journey.

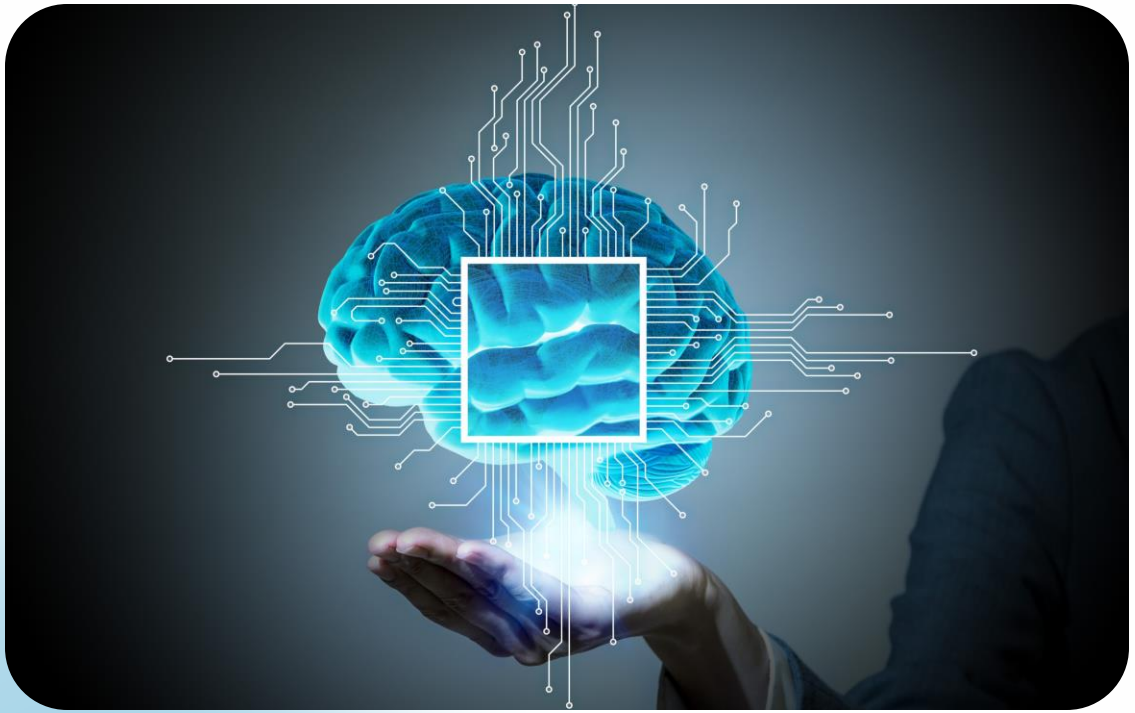
- How RNNs and LSTMs handle sequential data and model language structure
- How word embeddings and attention mechanisms capture meaning and relationships
- How the field transitioned from RNNs to Transformers and large Foundation Models
- Key terminology used in LLM development and evaluation
- How LLMs are trained, including pre-training, fine-tuning and reinforcement learning from human feedback (RLHF)
- The limitations, risks and future possibilities of LLMs

Topics Covered

- Understanding Neural Networks Using Analogy
- Backpropagation in Machine Learning
- Understanding Embeddings: The Library of Meaning
- Understanding Attention Mechanism
- ML in Real Life
- Tokens in AI
- Scaling Laws in AI

Understanding Neural Networks Using Analogy

Neural Networks: Inspired by the Brain

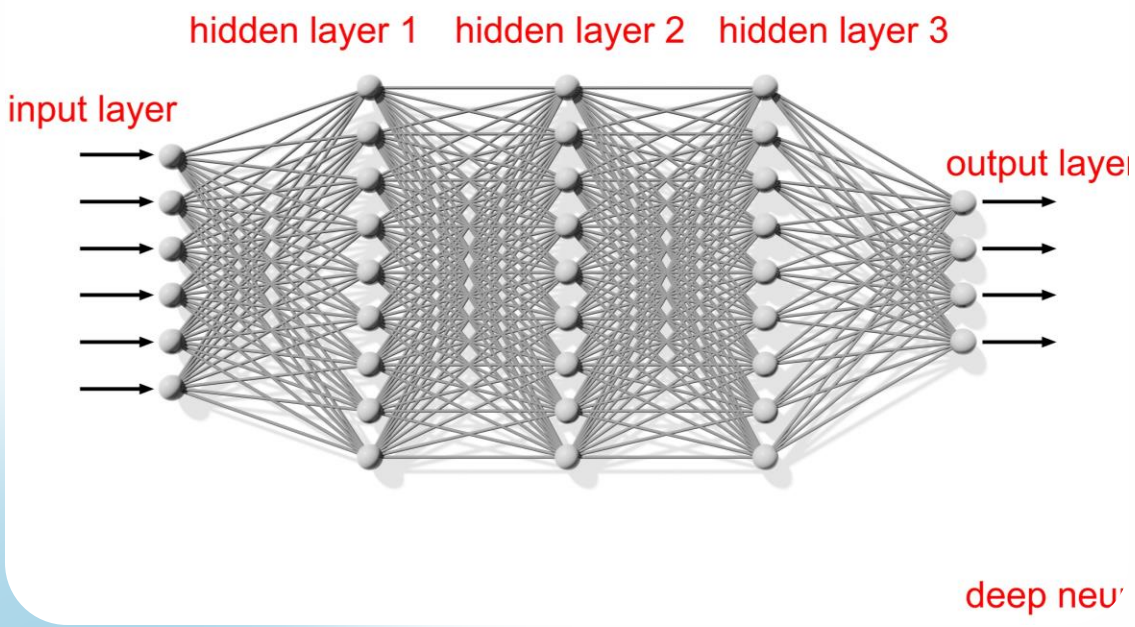


A neural network is a computational model that mirrors the structure and function of the human brain.

It comprises interconnected neurons in three layers, each with a key role in processing information.

Neural Networks: Inspired by the Brain

Key Components



Information flows from input to output via weighted connections, each reflecting the influence between neurons—akin to brain synapses.

Input layer
Receives raw data such as pixel values, text or sensor readings

Hidden layers
Perform complex transformations and extracts meaningful features from the input data

Output layer
Produces the final prediction or result based on the processed information

Learning by Adjusting Weights to Minimise Errors

Neural networks learn by adjusting weights between neurons, much like how the brain strengthens pathways through experience—rather than simply memorising data.

Forward pass
Network processes input data and generates predictions

Error calculation
Loss function compares predictions with correct answers

Backpropagation
Error signals propagate backward through the network

Weight adjustment
Connection weights update to reduce future errors

Learning by Adjusting Weights to Minimise Errors

The network gradually improves performance through iterative optimisation

It gains true power by generalising from training data to unseen cases — mimicking human learning through experience

Understanding Neural Networks: Classroom Analogy

Analogy: Imagine a neural network as a classroom of students collaborating to solve a problem.

The Students: Each neuron acts like a student, processing and passing information forward.

The Rows: Each layer forms a row of students, sharing notes to the next.

The Connections: Each connection/ weight defines how strongly one student listens to another — some voices carry more weight.

Understanding Neural Networks: Classroom Analogy

Key Points to Consider

The network improves over time by adjusting weights based on feedback, gradually reducing errors

It optimises accuracy by strengthening trusted connections, much like students learning whose input to rely on

AI Parameters: The Spices That Perfect the Recipe

Analogy: Imagine AI training as a master chef perfecting a complex dish.

- **Constant tasting**: The AI compares predictions to reality, identifying needed adjustments.
- **Billions of tiny knobs:** Each parameter acts like a fine adjustment, subtly refining the model.
- **Harmonious blend**: Cumulative tweaks produce accurate, fluent and well-calibrated results.

Like a chef's seasoning, the power lies in billions of precise changes working in harmony.

Backpropagation in Machine Learning

Backpropagation: Learning Like Cooking a New Dish

Analogy: Backpropagation in machine learning closely mirrors how we learn everyday skills — especially when trying a new recipe.

- **First attempt**: You follow a recipe and taste the result (neural network makes a prediction)
- **Spot the error**: You notice it's too salty or not spicy enough (this is the error detection)
- **Make adjustments**: You adjust ingredients - less salt, more spices (updating parameters)
- **Iterate until perfect**: You keep tasting and adjusting until the dish tastes right (reducing error)

Neural networks learn from mistakes by adjusting weights step by step

Backpropagation refines the internal "recipe" through trial and error, like a chef perfecting a dish

Understanding Embeddings: The Library of Meaning

Understanding Embeddings: The Library of Meaning

Analogy: Imagine Embeddings like a perfectly organised library, where each book is shelved by meaning.

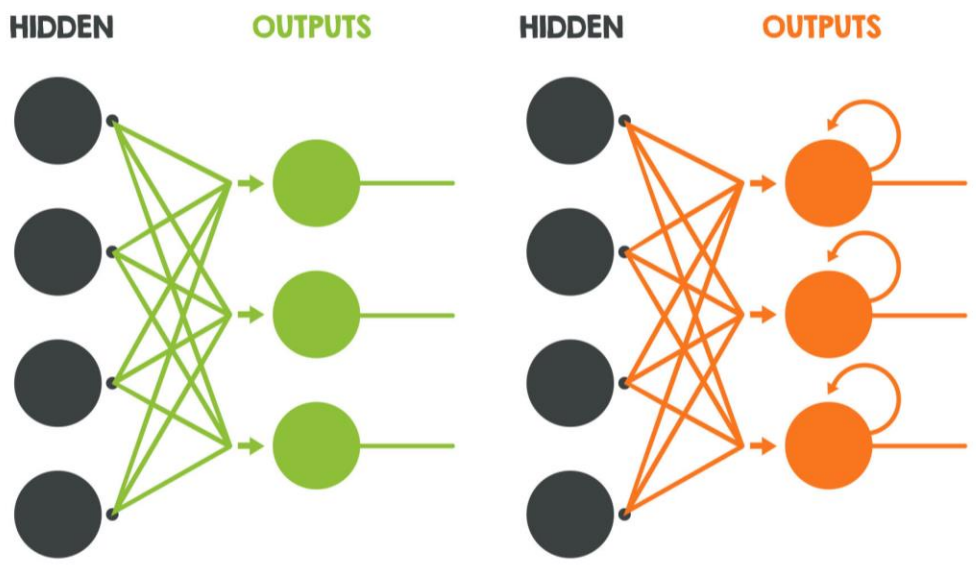
- Each book is assigned precise coordinates based on meaning
- Recipes, ingredients, and techniques cluster in the cooking section
- Sports books group by related activities — football near basketball, tennis near badminton
- Searching for "pizza" places it near "pasta" and "tomato sauce“, far from "astrophysics".

Embeddings cluster similar concepts in a mathematical space, like librarians shelving books by subject

In machine learning, these locations become numerical coordinates that help computers grasp concept relationships.

How RNNs Work Like Listening to a Story

Analogy: Imagine a friend sharing a long, winding vacation story, much like how Recurrent Neural Networks process sequences.



Step-by-step processing: You process each sentence sequentially, like an RNN handling one input at a time

Memory retention: Your brain retains key details to follow the story, like an RNN's hidden state

The Forgetting problem: After 20 minutes, recalling the beginning becomes difficult, like RNNs facing vanishing gradients over long sequences

How RNNs Work Like Listening to a Story

Key Points to Consider

Standard RNNs handle short-term dependencies but struggle with longer ones — like human memory

LSTMs and GRUs retain key details longer by adjusting weights — like taking notes during a story

Learning Like a Student: How LSTMs Work

Analogy: Studying for an exam mirrors how neural networks process and retain information.

- Regular RNNs forget earlier inputs, like students with short attention spans
- LSTMs use gates to manage memory, like students using notes and reminders
    - **Forget gate**: Removes unimportant information
    - **Input gate**: Decides what new information to store
    - **Output gate**: Selects what to recall when responding

LSTMs link past and present inputs, powering translation, speech and forecasting, until Transformers took over.

Understanding Attention Mechanism

Understanding Attention Mechanism

Analogy:

- **At a noisy party, your brain tunes into your name, filtering out distractions**
- **Attention mechanisms in AI do the same, focusing on key inputs while ignoring the rest**

Input processing
Your brain hears all voices, like AI handling every word in a sentence

Selective focus
You prioritise key information, such as your name over background noise

Meaningful connection
Focused attention helps you grasp what matters in context

Understanding Attention Mechanism

Key Points to Consider

Attention helps AI focus on key words like "cat" and "sat“, tuning out common ones, just as you hear your name in a noisy room.

This selective focus enables accurate translation, nuanced understanding and coherent generation

Traditional Models (RNNs) v. Transformer Models

Key features	Traditional models (RNNs)	Transformer models
Processing	**Sequential**: one input at a time	**Parallel**: all inputs simultaneously
Memory	**Limited**: early inputs easily forgotten	**Selective**: focus on important inputs
Learning	**Inefficient**: slow and hard to track information	**Efficient**: fast with better overall understanding

Self-attention in Transformers is like students spotting who matters most, weighing word relevance across the whole input, not just in order.

Understanding Self-Attention: How Words Work Together in Context

Analogy: At a party, hearing "by the bank" could mean a river or a financial institution — context is key.

You rely on surrounding context to interpret meaning:

- If the next words are "with a fishing rod" → it's probably a riverbank
- If you hear "withdrawing money" → it's definitely a financial bank

Understanding Self-Attention: How Words Work Together in Context

Self-attention lets each word focus on others to find its meaning in context

Like scanning a group chat, meaning shifts with surrounding messages — "That was cool" depends on the topic

ML in Real Life

Machine Learning Cooking Class: Parameters v. Hyperparameters

Parameter: The Flavour Profile
Analogy: Taste the flavour of the dish while cooking (saltiness, sweetness, spiciness)

- Adjusted automatically as the model trains
- The model "tastes" and tweaks these values itself
- Example: Your soup "learns" the perfect saltiness as ingredients interact

Hyperparameter: The Recipe Plan
Analogy: Taking the cooking decisions before starting (temperature, cooking time)

- Set by humans before training begins
- Control how the learning process happens
- Example: Choosing to bake at 180°C for 30 minutes vs. 200°C for 20 minutes

Parameters are learned during training, while hyperparameters are set beforehand.

ML in Real Life: How Training & Inference Work Like Studying for Exams

Training phase: The Study Marathon
Analogy: studying for months before exam

- Involves studying and solving problems
- Learns from mistakes, like backpropagation
- Demands time, energy, and computing power
- Often repeats tough concepts to improve understanding

Inference phase: Exam Time!
Analogy: Similar to taking the actual test

- Applies learned knowledge to new questions
- Works faster than training or studying
- Uses existing understanding — no further learning
- Delivers quick results from prior preparation

Remember: Training = Learning phase (slow, effortful, practice) while Inference = Application phase (fast, quick answers, real-world use).

Understanding Overfitting vs. Generalisation

The Cooking Analogy

Overfitting	Generalisation
Memorises one exact recipe, struggles with any variation	Learns core principles — adapts to new kitchens and ingredients
Fails outside familiar conditions — lacks adaptability	Applies flexible understanding — succeeds across scenarios

Understanding Overfitting vs. Generalisation

The Exam Preparation Analogy

Overfitting	Generalisation
Memorises exact answers, lacks true understanding	Masters core concepts, builds flexible understanding
Fails with new formats, limited to familiar scenarios	Handles varied formats, adapts to different question styles

The Key Difference

Overfitting = memorisation without understanding. Generalisation = conceptual learning that transfers to new situations.

Tokens in AI

What Are Tokens in AI?

- Tokens are small text units, like words or subwords, that the model reads
- AI splits words like "chatbot" into parts, e.g., "chat" and "bot" for processing

The Puzzle Analogy

- Break a sentence (the complete picture) into pieces (tokens) for understanding
- SoTreat some words as single pieces, e.g. “cat”
- - Split other words into multiple pieces, e.g. “chatbot” → “chat” + “bot”

What Are Tokens in AI?

The Pizza Analogy

- Treat a sentence as a whole pizza
- View tokens as individual slices
- Process one slice at a time to understand the full meal

What Are Tokens in AI?

Importance

AI uses tokens as language building blocks — like puzzle pieces forming a full picture

Token limits (e.g., 4,000 or 8,000) cap input size — understanding them helps manage AI processing

Scaling Laws in AI

Scaling Laws in AI: The Recipe for Better Performance

The Fundamental Rule of Thumb In AI Development:

More data + more parameters + more compute → better performance (to a point)

Analogy:

Like cooking a perfect dish, scaling key ingredients improves AI — but only to a point.

Scaling Laws in AI: The Recipe for Better Performance

Data = Ingredients

Use fresh, diverse data to enrich model performance

Parameters = Recipe Steps & Spices

Refine model architecture to enhance output quality

Compute = Kitchen Tools & Time

Apply powerful resources to improve processing efficiency

Balance is key: Scaling AI works best with just the right mix.

Q & A

Thank you